

Vehicle Management System

Java Abstract Class Assignment Report

Student Name: Ganesh Bhusare

Course: BE Artificial Intelligence & Data Science

College: Adsul's Technical Campus

Objective

To implement Object-Oriented Programming concepts such as Abstract Classes, Inheritance, Method Overriding, Constructors, and Polymorphism using a Vehicle Management System.

Problem Statement

Design a Vehicle Management System where different vehicles share common properties but calculate fuel efficiency and produce sounds differently.

Classes Used

Vehicle (Abstract Class): brand, speed, fuelCapacity, displayInfo(), calculateFuelEfficiency(), makeSound()

Car: Implements fuel efficiency calculation and sound generation.

Bike: Implements fuel efficiency calculation and sound generation.

Truck: Implements fuel efficiency calculation and sound generation.

ElectricCar: Additional subclass with custom implementation.

Main: Creates objects and displays outputs.

Concepts Covered

Abstract Class, Abstract Methods, Inheritance, Method Overriding, Polymorphism, Constructors, and ArrayList.

Conclusion

The Vehicle Management System demonstrates the practical implementation of Java Abstract Classes and other Object-Oriented Programming concepts. The project shows how abstraction, inheritance, and polymorphism can be used to create reusable, organized, and maintainable code.